AA200 - Applied Aerodynamics
Winter Quarter 2025-26
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PROBLEM SET 4 **Due Date: March 9th, 2026, 11:59 pm**

Solution of Laminar/Turbulent Flows Using Integral Boundary Layer Models

The objective of this problem set is to use the handout provided in canvas (with a Fortran code from Moran included) and write your own boundary layer solver that includes a laminar portion, a simple transition criterion (Michel's criterion), and a turbulent flow portion. For this purpose we will use both Thwaites' and Head's methods for the solution of the integral momentum equation for the boundary layer to obtain viscous solutions of the *attached* flow over an airfoil in incompressible flow. Notice that we will do this by assuming that the boundary layer *does not* influence the outer inviscid flow. This is only a two-step procedure:

1. Calculate the inviscid flow over the airfoil geometry at the specified angle of attack to obtain $V_e(x)$, $\frac{dV_e(x)}{dx}$, $\frac{dp}{dx}$ on the upper and lower surfaces of the airfoil.

2. Use the calculated $V_e(x)$, $\frac{dV_e(x)}{dx}$, $\frac{dp}{dx}$ to obtain distributions of H, θ, and δ^* along the upper and lower surfaces of the airfoil and use them to compute the coefficient of drag, C_d, of the airfoil.

No iteration between the outer inviscid flow and the boundary layer solution is required!!!. To accomplish this goal, complete the following steps / problems.

Problem 1. [60 points] *Development of Thwaites' and Head's Methods for Boundary Layer Solution.* Implement in Python, MATLAB (or the programming language of your choice, but you **cannot reuse directly the FORTRAN code provided**) the solution procedure for both Thwaites' and Head's methods and use them to obtain *numerical* solutions of the boundary layer on a flat plate with zero pressure gradient for a Reynolds number based on the length of the plate, $Re_L = 1 \times 10^6$. Provide the following information:

- Using Thwaites' method, and assuming that the boundary layer remains laminar throughout, calculate and plot the momentum and displacement thicknesses, θ and δ^*, the shape factor, H, and the local coefficient of skin friction, c_f. Integrate c_f and provide a number for the total coefficient of skin friction, C_f, for the plate. Compare all of these results to those from Blasius' solution.

- Using Head's method, and assuming that the boundary layer is tripped at the leading edge of the plate and remains turbulent throughout, calculate and plot the momentum and displacement thicknesses, θ and δ^*, the shape factor, H, and the local coefficient of skin friction, c_f. Integrate c_f and provide a number for the total coefficient of skin friction, C_f, for the plate.

Compare all of these results to those from Blasius' solution (to show differences between laminar and turbulent profiles) and to the simple power law estimates provided by Schlichting, which can be found in one of the class handouts.

Note: you can do anything you consider reasonable to deal with *undesirable* behavior at the leading edge of the flat plate.

Problem 2. [40 points] *Numerical Solution of the Viscous Flow Over a NACA 2410 Airfoil with a chord Reynolds number,* $Re_c = 5 \times 10^6$. Using the program in Problem 1, and your implementation of the Hess-Smith panel method construct a numerical procedure that follows the two steps explained above to calculate the boundary layers on the upper and lower surfaces of the airfoil using Michel's transition criterion at a single angle of attack of 1°. You can neglect the effects of curvature in your BL calculations: in other words, you can run the calculations for the BL as if the arclength distance along the upper/lower surfaces of the airfoil is laid out flat. In your discussion, address the following points:

- For the NACA 2410 airfoil at 1° angle of attack, compute and plot the following quantities for both the upper and lower surface boundary layers: θ, δ^*, H, c_f vs. distance from the leading edge stagnation point. Indicate the location of transition for both the upper and lower surfaces. You may use XFOIL results for guidance although the results will NOT be identical.
- Can you comment on the difficulties that you experience at the leading and trailing edges of the airfoil? What can be done to remove those? Can you explain what Moran's code does at the point of flow transition?

Please compute and plot your quantities with 160 panels generated with the `naca4.m` *function.*

Problem 3. Extra Credit. 15% *Inviscid-Viscous Coupling.* Using the code from Problem 2, compute the effect of the boundary layer on the inviscid outer flow with the same airfoil, Re_c and angle of attack. When the viscous effects are taken into account, the effective airfoil shape must be displaced perpendicularly by the displacement thickness δ^*. This provides a new inviscid shape from which the Hess-Smith code can be run, resulting in a new δ^*. After several iterations, the boundary layer will converge (but this is challenging and requires underrelaxation: ask us, if interested). In your discussion, address the following points:

- Plot the inviscid/viscous C_p and determine how much the c_l decreases? Is this comparable to that seen in XFOIL?
- What is the drag coefficient of the airfoil? Remember to take into account the varying dynamic pressure across the airfoil into the c_f calculation and don't forget the pressure drag.